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Chapter 18

Cisco IOS Fundamentals

In this chapter, you are exposed to questions that relate to fundamental operation of your Cisco device, such as understanding the boot process, understanding IOS images, managing files with IFS commands, and basic licensing operations.

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Understanding the Cisco IOS boot process
- Working with IOS images
- Managing Cisco IOS files
- Working with licenses

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- Your Cisco device has different types of memory: ROM memory stores the POST routines, bootstrap program, and the mini-IOSes, while flash memory stores the IOS images. RAM (or VRAM) stores temporary information such as your current configuration, MAC table, and routing table. Non-volatile RAM (or NVRAM) stores your startup configuration.
- Ensure you understand the boot process of a Cisco device. On startup, the POST routines are performed, then the bootstrap program locates an IOS in flash memory and extracts it to RAM. The system then checks for a startup configuration file; if one does not exist it will check for a TFTP server. If a TFTP server does not exist with the startup configuration then the device will prompt for the initial configuration dialog.
- Know that you can view the current IOS information with the `show version` command. If you wish to view all of the IOS images you have in flash memory you can use the show flash command. Be sure to understand how to read the show version output and identify information from the IOS details displayed with that command.

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Understanding the Cisco IOS Boot Process

738–750 Choose the best answer(s).

738. Where is the POST routines stored on a Cisco device?

(A) NVRAM

(B) RAM

(C) ROM

(D) Flash memory

739. Which of the following describes the benefit of booting from a TFTP server?

(A) If the server is unavailable then the backup IOS is used.

(B) A central place to upgrade the IOS

(C) Faster than loading from flash

(D) Slower than loading from flash

740. What happens after the bootstrap program locates the IOS in flash memory?

(A) It formats the IOS.

(B) It copies it to ROM.

(C) The IOS is loaded into RAM.

(D) It copies it to NVRAM.

741. You wish to disrupt the boot process on the Cisco device. What keystroke would you press?

(A) Ctrl-Alt-Break

(B) Ctrl–Break

(C) Ctrl-Alt-Ins

(D) Alt-Break

742. Which type of memory is used to store the bootstrap program?

(A) NVRAM

(B) RAM

(C) ROM

(D) Flash memory

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743. The startup configuration is stored in which type of memory?

(A) ROM

(B) NVRAM

(C) RAM

(D) FLASH

744. What happens if a startup config does not exist on boot-up?

(A) The device will try to connect to a TFTP server for the configuration.

(B) The device fails to boot.

(C) The device loads a default config from ROM.

(D) The device loads a default config from flash.

745. What command would you use to configure the router to not load the startup configuration?

(A) `config-register 2102`

(B) `config-register 1200`

(C) `config-register 2100`

(D) `config-register 2142`

746. Which component is responsible for locating an operating system to load in memory?

(A) Bootstrap

(B) ROM

(C) Flash

(D) IOS

747. Looking at the figure below, what is the command to configure the Cisco device to boot from the IOS stored in flash memory?

```
R1#show flash

System flash directory:
File  Length   Name/status
  3   50938004 c2800nm-advipservicesk9-mz.124-15.T1.bin
  2   28282    sigdef-category.xml
  1   227537   sigdef-default.xml
[51193823 bytes used, 12822561 available, 64016384 total]
63488K bytes of processor board System flash (Read/Write)

R1#
```

(A) `boot system tftp c2800nm-advipservicesk9-mz.124-15.T1.bin`

(B) `boot system flash c2800nm-advipservicesk9-mz.124-15.T1.bin`

(C) `boot system flash sigdef-category.xml`

(D) `boot system flash sigdef-default.xml`

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748. What command would you use to change the configuration file that is applied on startup?

(A) `boot config <file>`

(B) `set startup-file`

(C) `boot system config <file>`

(D) `boot private-config <file>`

749. You are troubleshooting the startup of your Cisco switch. What command was used in the figure below to display the following output?

```
BOOT path-list      :
Config file         : flash:/config.text
Private Config file : flash:/private-config.text
Enable Break        : no
Manual Boot         : no
HELPER path-list    :
Auto upgrade        : yes
NVRAM/Config file
      buffer size:   65536
Switch#
```

(A) `show config`

(B) `show IOS`

(C) `show flash`

(D) `show boot`

750. The bootstrap program is stored in what type of memory on the Cisco router?

(A) NVRAM

(B) RAM

(C) ROM

(D) Flash memory

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Working with IOS Images

751–760 Choose the best answer(s).

751. Looking at the figure below, what is the name of the image file that contains the Cisco IOS?

```
R1#show flash

System flash directory:
File  Length   Name/status
  3   50938004 c2800nm-advipservicesk9-mz.124-15.T1.bin
  2   28282    sigdef-category.xml
  1   227537   sigdef-default.xml
[51193823 bytes used, 12822561 available, 64016384 total]
63488K bytes of processor board System flash (Read/Write)

R1#
```

(A) sigdef-default.xml

(B) sigdef-category.xml

(C) name/status

(D) c2800nm-advipservicesk9-mz.124-15.T1.bin

752. You wish to back up your IOS image to a TFTP server. What command would you use?

(A) `copy flash tftp`

(B) `copy boot tftp`

(C) `copy image tftp`

(D) `copy <filename> tftp`

753. Which of the following would be a downfall of loading the IOS image from a TFTP server?

(A) If the server is unavailable devices cannot boot up.

(B) Faster than flash

(C) A central place to upgrade the IOS

(D) Fault tolerance available

754. You have a copy of your IOS image located on a TFTP server. What command would you use to restore the IOS to your device?

(A) `restore flash tftp`

(B) `restore tftp flash`

(C) `copy tftp rom`

(D) `copy tftp flash`

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755. You wish to find out what version of the Cisco IOS you are running. What command would you run?

(A) `show mem`

(B) `show memory`

(C) `show ios`

(D) `show version`

756. What command would you use to configure the Cisco device to boot from an IOS image located on a TFTP server?

(A) `boot tftp <filename.bin>`

(B) `boot system <filename.bin>`

(C) `boot system tftp <filename.bin>`

(D) `boot system flash <filename.bin>`

757. By default, the Cisco bootstrap program loads which IOS image if one is not specified?

(A) The first image on a TFTP server

(B) The last image on a TFTP server

(C) The last image stored in flash

(D) The first image stored in flash

758. What command would you use to specify a secondary bootstrap image?

(A) `boot config`

(B) `boot system`

(C) `boot bootstrap`

(D) `boot`

759. You need to find out what IOS image file was used to boot the router. What command would you use?

(A) `show ios`

(B) `show version`

(C) `show boot`

(D) `show image`

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760. What command would you use to configure your router to load the IOS image from a TFTP server?

(A) `boot system flash <filename>`

(B) `boot system <filename>`

(C) `boot system tftp <filename>`

(D) `boot system flash-tftp <filename>`

Managing Cisco IOS Files

761–772 Choose the best answer(s).

761. You wish to delete a file from flash memory. What command would you use?

(A) `delete flash:<filename>`

(B) `kill flash:<filename>`

(C) `delete <filename>`

(D) `kill <filename>`

762. You wish to invoke the initial configuration dialog. What command would you use?

(A) `init`

(B) `show init`

(C) `init dialog`

(D) `setup`

763. What file extension do Cisco IOS images use?

(A) .jpg

(B) .image

(C) .exe

(D) .bin

764. You are customizing the boot preferences and would like to change the size of the NVRAM buffer used for the IFS. What command would you use?

(A) `boot buffersize`

(B) `boot config-file`

(C) `buffersize-boot`

(D) `config-file boot`

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765. You would like to know how much memory exists on the Cisco device. What command would you use?

(A) `show memory`

(B) `show version`

(C) `show ram`

(D) `show ios`

766. Looking at the output of the figure below, what command was used?

```
Directory of flash:/

    1  -rw-     4414921          <no date>  c2960-lanbase-mz.122-25.FX.bin
    2  -rw-         676          <no date>  vlan.dat

64016384 bytes total (59600787 bytes free)
Switch#
```

(A) `show system:`

(B) `show flash:`

(C) `show version`

(D) `show nvram`

767. What command can you use on your Cisco router to display the available flash memory?

(A) `show version`

(B) `show memory`

(C) `show flash`

(D) `show mem`

768. You wish to display the contents of NVRAM. What command would you use?

(A) `dir nvram:`

(B) `dir`

(C) `dir flash:`

(D) `show nvram`

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769. Looking at the figure below, what is the platform for the device?

```
R1#show version
Cisco IOS Software, 2800 Software (C2800NM-ADVIPSERVICESK9-M), Version 12.4(15)T1,
RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 06:21 by pt_rel_team

ROM: System Bootstrap, Version 12.1(3r)T2, RELEASE SOFTWARE (fc1)
Copyright (c) 2000 by cisco Systems, Inc.

System returned to ROM by power-on
System image file is "c2800nm-advipservicesk9-mz.124-15.T1.bin"

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
M860 processor: part number 0, mask 49
2 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
239K bytes of NVRAM.
62720K bytes of processor board System flash (Read/Write)

Configuration register is 0x2102

R1#
```

(A) 2800

(B) 12.4

(C) 2 FastEthernet

(D) 2102

770. Which file contained in flash memory on a Cisco switch is used to store secured configuration data such as cryptography keys?

(A) config.text

(B) secured.text

(C) keys.text

(D) private-config.text

771. You wish to display the contents of flash memory using IFS commands. What command would you use?

(A) `dir nvram`

(B) `dir flash:`

(C) `display flash`

(D) `get flash:`

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772. Looking at the figure below, how much flash memory exists in the device?

```
R1#show version
Cisco IOS Software, 2800 Software (C2800NM-ADVIPSERVICESK9-M), Version 12.4(15)T1,
RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 06:21 by pt_rel_team

ROM: System Bootstrap, Version 12.1(3r)T2, RELEASE SOFTWARE (fc1)
Copyright (c) 2000 by cisco Systems, Inc.

System returned to ROM by power-on
System image file is "c2800nm-advipservicesk9-mz.124-15.T1.bin"

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
M860 processor: part number 0, mask 49
2 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
239K bytes of NVRAM.
62720K bytes of processor board System flash (Read/Write)

Configuration register is 0x2102

R1#
```

(A) 62720 K

(B) 2800 K

(C) 239 K

(D) 5120 K

Working with Licenses

773–775 Choose the best answer(s).

773. You wish to remove a license from your Cisco device. What command would you use?

(A) `remove license ipservices`

(B) `delete ipservices`

(C) `license clear ipservices`

(D) `delete license ipservices`

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774. You have copied a new Cisco license file called r1fs-ips from the TFTP server to flash memory. How do you install this new license on your Cisco switch?

(A) `license install tftp: r1fs-ips`

(B) `license install flash: r1fs-ips`

(C) `license rlfs-ips`

(D) `license flash: rlfs-ips`

775. You wish to display the current license information. What command would you use?

(A) `display license file`

(B) `show license file`

(C) `license show file`

(D) `show all`